

Q1 - 25 July - Shift 1

To increase the resonant frequency in series LCR circuit,

- (A) Source frequency should be increased
- (B) Another resistance should be added in series with the first resistance.
- (C) Another capacitor should be added in series with the first capacitor
- (D) The source frequency should be decreased

Space for your notes:

Q2 - 25 July - Shift 1

The rms value of conduction current in a parallel plate capacitor is $6.9 \mu\text{A}$. The capacity of this capacitor, if it is connected to 230 V ac supply with an angular frequency of 600 rad/s , will be :

- (A) 5 pF
- (B) 50 pF
- (C) 100 pF
- (D) 200 pF

Space for your notes:

Q3 - 25 July - Shift 2

When you walk through a metal detector carrying a metal object in your pocket, it raises an alarm. This phenomenon works on

- (A) Electromagnetic induction
- (B) Resonance in ac circuits
- (C) Mutual induction in ac circuits
- (D) interference of electromagnetic waves

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Q4 - 26 July - Shift 1

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Questions

MathonGo

In a series LR circuit $X_L = R$ and power factor of the circuit is P_1 . When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes P_2 . The ratio $\frac{P_1}{P_2}$ is

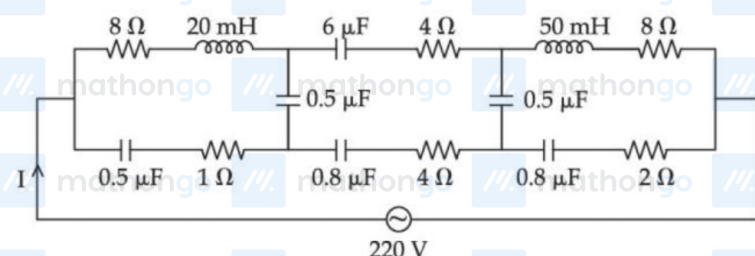
- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$
(C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $2 : 1$

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Q5 - 26 July - Shift 1

The effective current I in the given circuit at very high frequencies will be _____ A

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Q6 - 27 July - Shift 1

To light, a 50 W, 100 V lamp is connected, in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}} \mu\text{F}$, with 200 V, 50Hz AC source. The value of x will be _____.

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Q7 - 27 July - Shift 2

A series LCR circuit has $L = 0.01 \text{ H}$, $R = 10 \Omega$ and $C = 1 \mu\text{F}$ and it is connected to ac voltage of amplitude (V_m) 50 V . At frequency 60% lower than resonant frequency, the amplitude of current will be approximately :

- (A) 466 mA (B) 312 mA
(C) 238 mA (D) 196 mA

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Q8 - 28 July - Shift 1

The equation of current in a purely inductive circuit is $5\sin(49\pi t - 30^\circ)$. If the inductance is 30 mH then the equation for the voltage across the inductor, will be :

- $\left\{ \text{Let } \pi = \frac{22}{7} \right\}$
(A) $1.47\sin(49\pi t - 30^\circ)$ (B) $1.47\sin(49\pi t + 60^\circ)$
(C) $23.1\sin(49\pi t - 30^\circ)$ (D) $23.1\sin(49\pi t + 60^\circ)$

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Q9 - 28 July - Shift 1

The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1} . The value of resistance in the circuit is $R = 5\Omega$. The self inductance in the circuit is _____ mH.

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Q10 - 29 July - Shift 1

An alternating emf $E = 440 \sin 100\pi t$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}$ H. If an a.c. ammeter is connected in the circuit, its reading will be :

- (A) 4.4 A (B) 1.55 A
(C) 2.2 A (D) 3.11 A

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Q11 - 29 July - Shift 1

A coil of inductance 1 H and resistance 100 Ω is connected to a battery of 6 V. Determine approximately :

- (a) The time elapsed before the current acquires half of its steady – state value
(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given $\ln 2 = 0.693$,

$e^{-3/2} = 0.25$)

- (A) $t = 10$ ms; $U = 2$ mJ
(B) $t = 10$ ms; $U = 1$ mJ
(C) $t = 7$ ms; $U = 1$ mJ
(D) $t = 7$ ms; $U = 2$ mJ

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Q12 - 29 July - Shift 2

A circuit element X when connected to an a.c. supply of peak voltage 100 V gives a peak current of 5 A which is in phase with the voltage. A second element Y when connected to the same a.c. supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$. If X and Y are connected in series to the same supply, what will be the rms value of the current in ampere ?

- (A) $\frac{10}{\sqrt{2}}$ (B) $\frac{5}{\sqrt{2}}$ (C) $5\sqrt{2}$ (D) $\frac{5}{2}$

Space for your notes:

Q13 - 29 July - Shift 2

A capacitor of capacitance 500 μF is charged completely using a dc supply of 100 V. It is now connected to an inductor of inductance 50 mH to form an LC circuit. The maximum current in LC circuit will be _____ A.

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Answer Key

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Q1 (C)

Q2 (B)

Q3 (B)

Q4 (B)

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Q5 (44)

Q6 (3)

Q7 (C)

Q8 (D)

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Q9 (250)

Q10 (C)

Q11 (C)

Q12 (D)

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Q13 (10)

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Q1 (C)

$$f = \frac{1}{2\pi\sqrt{LC}}$$

To increase the resonating frequency product of L and C should decrease.

By joining capacitor in series, capacitor will decrease

Q2 (B)

FROM THIS: (B)

Current in capacitor $I = \frac{V}{X_C}$

$$I = (V) \times (\omega C)$$

$$C = \frac{I}{V\omega} = \frac{6.9 \times 10^{-6}}{230 \times 600} = 50 \text{pF}$$

Q3 (B)

Metal detector works on the principle of transmitting an electromagnetic signal and analyses a return signal from the target. So it works on the principle of resonance in AC circuit.

Q4 (B)

In case of L-R circuit

$$Z = \sqrt{X_L^2 + R^2} \text{ \& power factor}$$

$$P_1 = \cos \phi = \frac{R}{Z}$$

As $X_L = R$

$$\Rightarrow Z = \sqrt{2}R$$

$$\Rightarrow P_1 = \frac{R}{\sqrt{2}R} \Rightarrow P_1 = \frac{1}{\sqrt{2}}$$

In case of L-C-R circuit

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

As $X_L = X_C$

$$\Rightarrow Z = R$$

$$\Rightarrow P_2 = \cos \phi = \frac{R}{R} = 1$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{1}{\sqrt{2}}$$

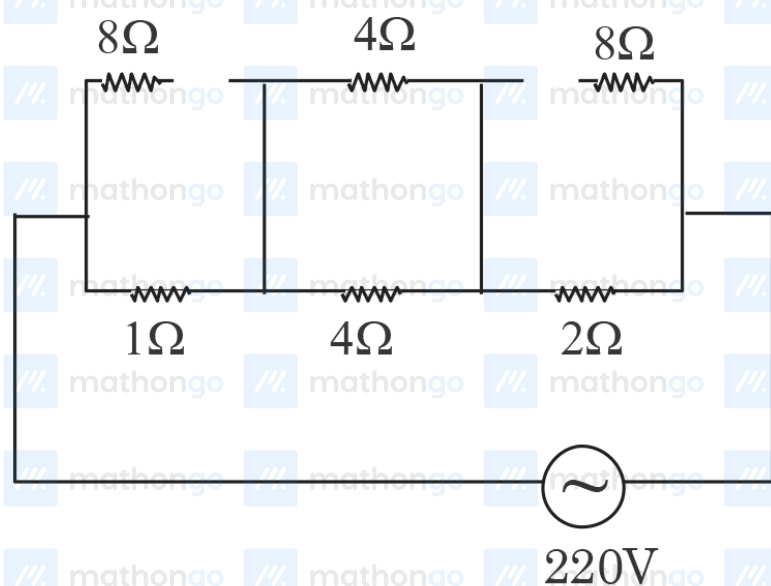
Q5 (44)

At very high frequencies,

$$X_C = \frac{1}{\omega C} \approx 0$$

$$\text{Also } X_L = \omega L \approx \infty$$

Thus, equivalent circuit can be redrawn as



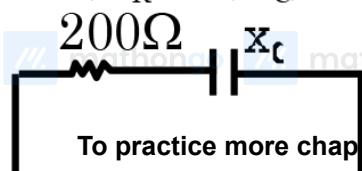
$$Z = 1 + 2 + 2 = 5\Omega$$

$$I = \frac{220V}{5\Omega} = 44A$$

Q6 (3)

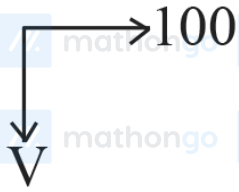
$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$$

$$(V_R) \quad (V_C)$$



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200 V, 50 Hz



$$R = \frac{100 \times 10^2}{50} = R = 200 \Omega$$

$$V_R^2 + V_C^2 = V^2$$

$$(100)^2 + V_C^2 = (200)^2$$

$$i = \frac{100}{200} = \frac{1}{2}; \quad V^2 = 40000$$

$$V = I \times X_C; \quad V_C^2 = 30000$$

$$V_C = 100\sqrt{3}$$

$$X_C = 200\sqrt{3}$$

$$200\sqrt{3} = \frac{1}{\omega C}$$

$$C = \frac{1}{20 \times 50 \times 20\sqrt{3}} = \frac{50 \times 10^{-6}}{\sqrt{3}}$$

$$C = \frac{50 \times 10^{-6}}{\sqrt{3}} \text{ F}$$

Q7 (C)

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$$\text{Resonant frequency, } \omega_0 = \frac{1}{\sqrt{LC}} = 10^4 \text{ rad/sec}$$

$$\omega' = .4 \times 10^4 = 4000 \text{ rad/sec}$$

$$i_0 = \frac{V_0}{\sqrt{R^2 + (X'_C - X'_L)^2}} = 238 \text{ mA}$$

Q8 (D)

$$V_0 = i_0 X_L$$

$$= i_0 (\omega L)$$

$$= (5)(49\pi)(30 \times 10^{-3})$$

$$= 23.1$$

Voltage will lead current by 90° .

$$\therefore V = 23.1 \sin (49 \pi t + 60^\circ)$$

Q9 (250)

$$\text{Band width} = 232 - 212 = \frac{R}{L}$$

$$\therefore L = \frac{5}{20} = \boxed{250} \text{ mH}$$

Q10 (C)

$$E = 440 \sin 100\pi t, L = \frac{\sqrt{2}}{\pi} \text{ H}$$

$$X_L = \omega L = 100 \pi \frac{\sqrt{2}}{\pi} = 100\sqrt{2} \Omega$$

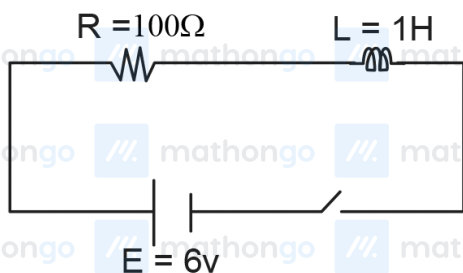
$$\text{Peak current } I_0 = \frac{E_0}{X_L} = \frac{440}{100\sqrt{2}} = 2.2\sqrt{2} \text{ A}$$

AC ammeter reads RMS value therefore reading will be I_{rms}

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = 2.2 \text{ A}$$

Q11 (C)

Given circuit is R – L growth circuit



$$i = \frac{E}{R}(1 - e^{-t/\tau})$$

$$i = \frac{E}{2R} = \frac{E}{R}(1 - e^{-t/\tau})$$

Solving $t = \tau \ln 2$

$$t = \frac{1}{R} \ln 2 = \frac{1}{100} 0.693 = 0.00693$$

$$= 7 \text{ ms}$$

$$i(15\text{ms}) = \frac{E}{R}(1 - e^{-\frac{15}{10}})$$

$$i = \frac{6}{100}(1 - 1/4) = \frac{3}{4} \times \frac{6}{100}$$

$$U = \frac{1}{2} LI^2,$$

by solving we get $U = 1 \text{ mJ}$.

Q12 (D)

Element X should be resistive with $R = 20\Omega$

Element Y should be inductive with $X_L = 20\Omega$

When X and Y are connector in series

$$Z = \sqrt{X_L^2 + R^2} = 20\sqrt{2}$$

$$I_0 = \frac{E_0}{Z} = \frac{100}{20\sqrt{2}} = \frac{5}{\sqrt{2}} \text{ A}$$

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = \frac{5}{2} \text{ A}$$

Q13 (10)

Energy stored in capacitor

$$= \frac{1}{2} CV^2 = \frac{1}{2} 500 \times 10^{-6} \times 10^4$$

$$= \frac{5}{2} \text{ J}$$

Current will be maximum when whole energy of capacitor becomes energy of inductor.

$$\frac{1}{2} LI^2 = \frac{5}{2}$$

$$I = \sqrt{\frac{5}{L}} = \sqrt{\frac{5}{50 \times 10^{-3}}} = 10 \text{ A.}$$